
Flood Origin Analysis of the Maelbeek Valley: An Integrated Study of the Summer 2024 Extreme Rain

2.2.2 DATA CALCULATION FOR SWMM MODEL

I. Junctions

- i. Invert elevation determined from the lowest connected pipe end (offset = 0).
- ii. Offsets for other pipes assigned based on their pipe end elevations.
- iii. For known areas (e.g., Rue Gray), invert elevations were directly available.
- iv. Junction coordinates identified from pipe endpoint locations.
- v. Maximum depth = surface elevation – invert elevation.
- vi. Initial depth, surcharge depth, and ponding depth assumed zero.

II. Conduits (Pipes)

- i. Most pipe properties (shape, length, elevations, geometry) were available.
- ii. Missing data estimated using the continuity principle (smooth connection between pipes).
- iii. Pipe ends adjusted to ensure consistent elevation and dimensions.
- iv. Pipes initially assumed to be dry (no initial flow).

III. Sub-catchments

- i. Defined as rooftops and surface areas.
- ii. Slopes obtained from GRASS GIS catchment analysis.
- iii. Rooftops assumed 100% impervious.
- iv. Impervious/pervious ratio derived from land use raster data.
- v. Green areas (grass, trees, fields) considered pervious.
- vi. Built surfaces (roads, concrete) considered impervious.
- vii. Areas calculated using GIS-based sub-catchment polygons.
- viii. Flow length estimated as:
 - a. Rooftops → $\sqrt{\text{area}}$
 - b. Surfaces → longest flow path approximation

2.2.3 SWMM MODEL INTEGRATION AND SIMULATION WORKFLOW

- i. Network Integration: We merged the unconnected sewer networks from two specific sub catchments into the main system.
- ii. Data Formatting: Using the existing main model as a base, we created a new .inp file with the required junctions and nodes by copying the features as required by SWMM format with the help of notepad+.
- iii. Sub catchment Consolidation: A separate .inp file was generated for the sub catchments and then merged with the junction and node data to create one unified model. This was completed by copying and pasting the new features into the sub catchment inp file in correct SWMM formats using Notepad++ to ensure data integrity.

- iv. Precipitation Setup: Rainfall data for gauges P14 and P11 was downloaded from Hydria; the original .tsv files were converted to .dat format for SWMM compatibility.
- v. Inflow Calculation: Data from collectors C10 and C11 were integrated to account for inflows from neglected sections, simplifying the network logic.
- vi. Hydraulic Connections: We established connections between main three pipes at various locations by introducing weirs into the pathways to manage flow transitions.
- vii. Event Analysis: Once the single functional model was generated, it was simulated and analyzed across four distinct rainfall events.

2.3 GROUNDWATER MODELING (MODELMUSE/MODFLOW-6)

To evaluate the subsurface hydrological dynamics of the Maelbeek Valley, a numerical groundwater model was constructed using the ModelMuse interface to run a MODFLOW-6 simulation. This model specifically aimed to determine if anthropogenic barriers and high-intensity rainfall contribute to groundwater emergence in Rue Gray.

2.3.1 MODEL SETUP AND BOUNDARY CONDITIONS

The model represents the "Sand of Brussels" aquifer, a confined sand bank resting on a primary clay layer.

- I. Hydraulic Properties: An initial horizontal hydraulic conductivity (K) of 17.28 m/d was assigned based on the Brussels Phreatic System Model v1.0. For the vertical conductivity, an anisotropy ratio of 0.1 (10% of K) was assumed.
- II. Boundary Conditions:
 - a. External: No-flow boundaries were defined at the topographical divides (South and East), while the Senne River (North and West) was implemented using the RIV package.
 - b. Internal Drainage: The urban sewer network was simulated using the DRN (Drain) package to account for groundwater baseflow into the collector system.
 - c. Infrastructure Barriers: The local railway and bridge foundations were digitized as impermeable vertical barriers (No-flow/Low-K) extending 10 meters deep to simulate their impact on lateral flow.

2.3.2 RECHARGE CALCULATION (CURVE NUMBER METHOD)

To simulate the specific storm event of July 9th, 2024, a spatially weighted Curve Number (CN) was calculated. Using land-use data (paved surfaces, buildings, and green spaces) and HEC-HMS reference tables, an equivalent CN of 82.86 was determined. Based on the FLOWBRU peak rainfall of 42.2 mm/day, the event-specific recharge was calculated at 0.0051 m/d.

Land Use and surface characteristics	Area (m ²)	CN
Agriculture	6536	81
forest	6266311	66
grass	4964275	79
buildings	14457395.7	85
paved	5356503	98
water	236390	100
roads	7971778	84

Table 1: Values of CN for each surface in the study area (HEC-HMS Technical Reference

Manual)

2.3.3 SIMULATION SCENARIOS (RUNS 1–5)

The modeling phase was executed through five distinct scenarios, progressively refining the model's ability to replicate the complex urban hydrogeology of the Rue Gray area.

- I. Run 1: Initial Steady-State Baseline
The primary goal was to establish a functional model using regional parameters from the Brussels Phreatic System Model v1.0. This run assumed a uniform horizontal hydraulic conductivity (K) of 17.28 m/d. Boundary conditions for the Senne River and the internal sewer drainage (DRN package) were set to represent long-term average conditions. This provided a benchmark to assess how much the local Rue Gray conditions deviated from regional averages.
- II. Run 2: Automated Calibration (PEST Implementation)
Recognizing that regional K values may not capture the localized compaction or siltation in the valley bottom, we utilized PEST (Parameter ESTimation) software. PEST iteratively adjusted the hydraulic conductivity to minimize the residuals between the model's predicted heads and 13 historical observation points. This resulted in an optimized horizontal K of 3.697 m/d, which significantly improved the model's accuracy and provided a reliable steady-state foundation.
- III. Run 3: Event-Based Response Simulation (July 9, 2024)
In this run, the calibrated steady-state model was subjected to the extreme conditions of the July 2024 storm. The recharge was increased to 0.0051 m/d (calculated via the Curve Number method). The objective was to observe the "mounding" of the water table. By comparing these results to the ground surface elevation, we could identify areas where the hydraulic head predicted groundwater emergence at the street level.
- IV. Run 4: Barrier Sensitivity and Structural Permeability

Initially, anthropogenic structures (the railway and bridge foundations) were modeled as perfectly impermeable barriers. In Run 4, we tested the sensitivity of the model by assigning a low but non-zero hydraulic conductivity to these structures. This allowed us to observe how sensitive the upstream "ground-water damming" effect is to the exact permeability of the urban infrastructure, providing a more nuanced view of subsurface flow bypass.

V. Run 5: Zero-Barrier Control Scenario

The final run involved the complete removal of the railway and building foundation barriers from the ModelMuse environment. By comparing the results of Run 5 against Run 2, we were able to isolate the "anthropogenic impact." This allowed the research to quantify exactly how much the water table rise is caused by natural valley topography versus how much is exacerbated by man-made obstacles blocking the natural drainage path toward the Senne.

3. RESULT AND DISCUSSION

The following section presents the findings derived from the integration of citizen-sourced data and numerical simulations. By synthesizing the empirical evidence from local residents with high-fidelity modeling outputs, we can diagnose the multi-faceted nature of the flooding in Rue Gray. The results are structured to first evaluate the visible surface phenomena through spatiotemporal mapping, followed by an in-depth analysis of the subterranean hydraulic and hydrogeological drivers. This comparative approach allows for a clear distinction between sewer-driven surcharges and groundwater-driven emergence, providing a comprehensive overview of the "compound" flood risk characterizing the study area.

3.1 SPATIOTEMPORAL INTERPRETATION OF CITIZEN OBSERVATIONS

The synthesis of citizen reports and high-resolution rainfall data provided a clear diagnostic of the flood dynamics during the summer 2024 season. By analyzing the resulting heatmaps and rainfall time series, several key patterns emerged:

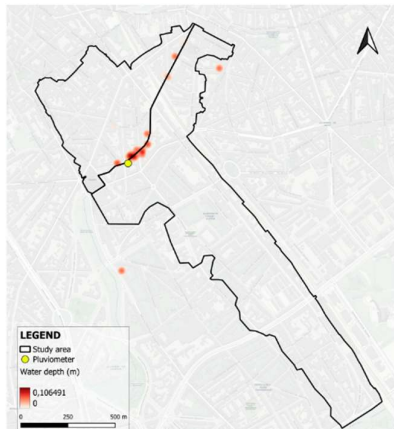


Figure 2: Heatmap 1. event (9.7.2024)

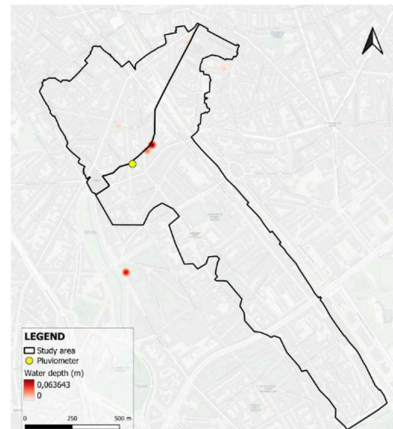


Figure 3: Heatmap 2. event (12.7.2024)

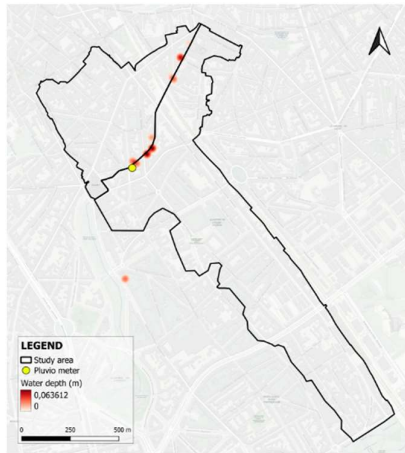


Figure 4: Heatmap 3. event (1.8.2024)

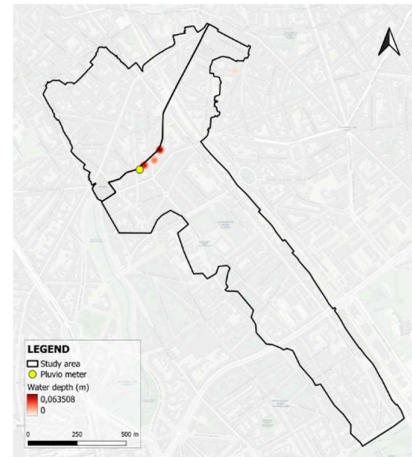


Figure 5: Heatmap 4. event (14.8.2024)

The heatmaps (Figures 2–5) reveal a consistent concentration of flooding in the lowest topographical points of the Maelbeek Valley, specifically centered around the Rue Gray corridor. Regardless of the varying rainfall intensities across the four events, the "flood-prone" clusters remained geographically stable. This suggests that the street's design and its position at the valley bottom act as a primary physical constraint, naturally collecting runoff from the surrounding steep slopes.

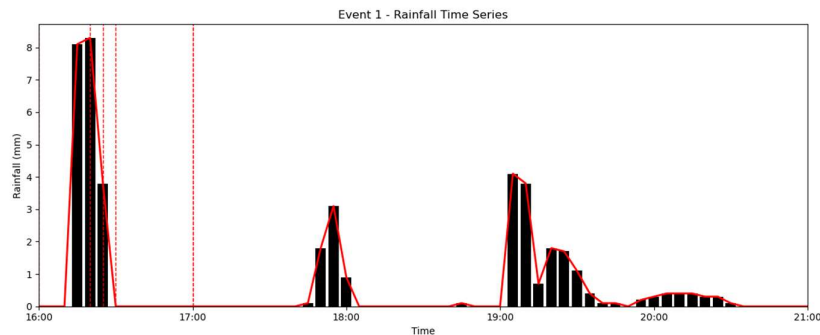


Figure 6: Rainfall Time Series 1. Event (9.7.2024)

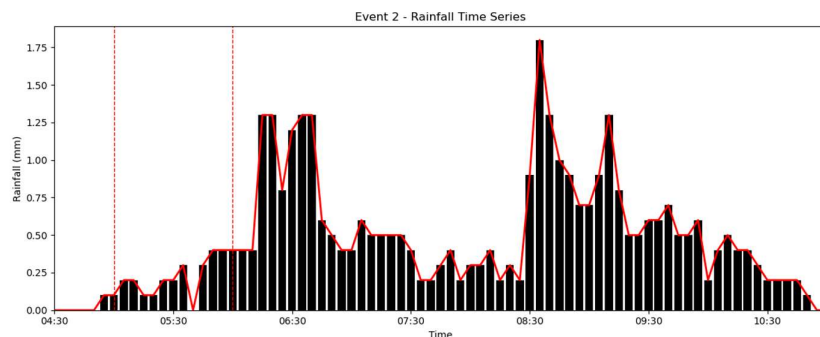


Figure 7: Rainfall Time Series 2. Event (14.7.2024)

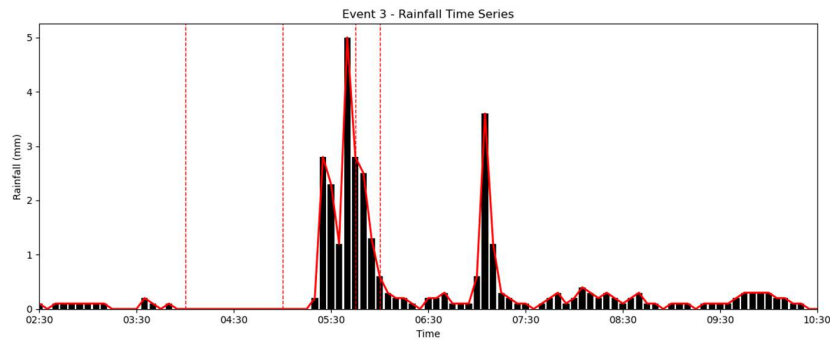


Figure 8: Figure 8 Rainfall Time Series 3, Event (1.8.2024)

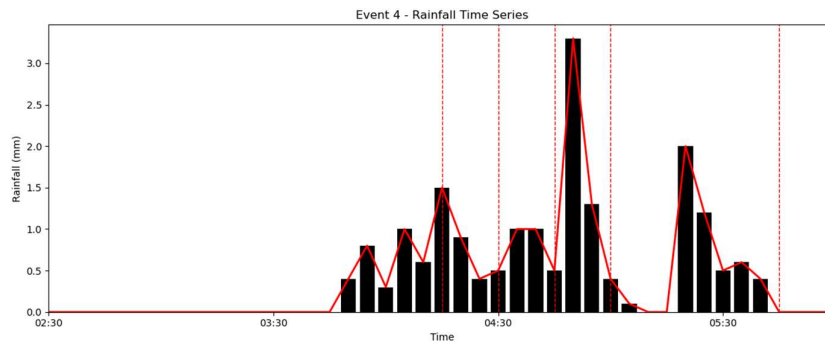


Figure 9: Rainfall Time Series 4, Event (14.8.2024)

A comparison between the rainfalls peaks (Figures 6–9) and the timestamps of citizen reports indicates an extremely rapid response time. In most instances, flooding was reported within minutes of peak precipitation. This immediate reaction is a classic indicator of pluvial flooding and sewer surcharge (“brown water”), rather than groundwater rise, which typically exhibits a delayed response.

3.2 EVALUATION OF DRAINAGE DYNAMICS

As illustrated in Figure 10, the study area was divided into high-resolution subcatchments to accurately represent the urban fabric of the Maelbeek Valley. The mapping (shown in orange and green) highlights the high percentage of impervious surfaces—buildings and paved roads—which facilitate near-instantaneous runoff during rainfall. The blue line traces the primary collector path, which follows the natural topographic low-point of the valley, funneling all upstream discharge directly toward the Rue Gray corridor.

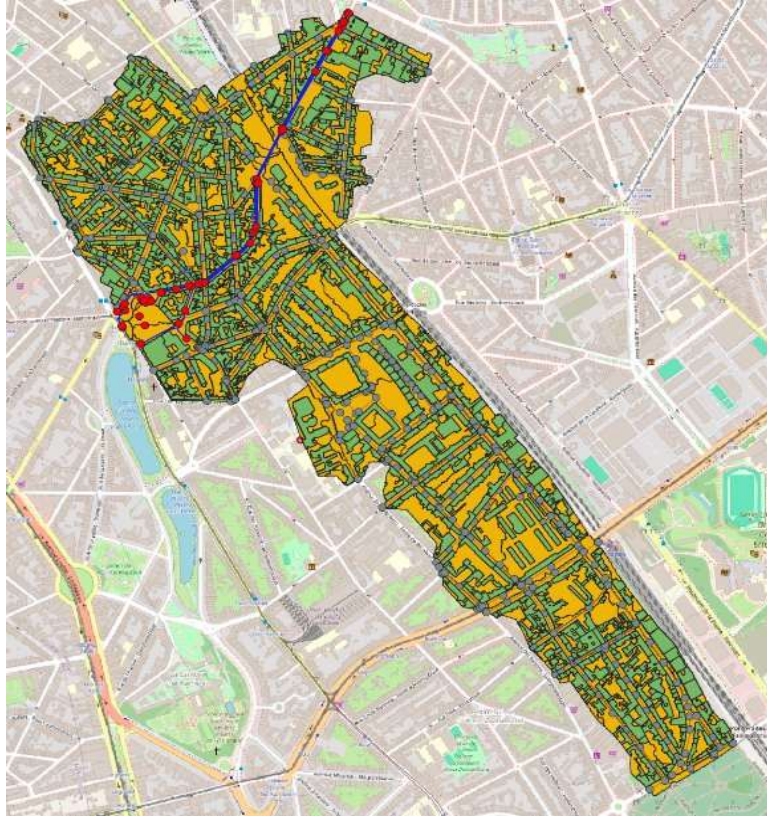


Figure 10: Subcatchment Delineation and Surface Characteristics

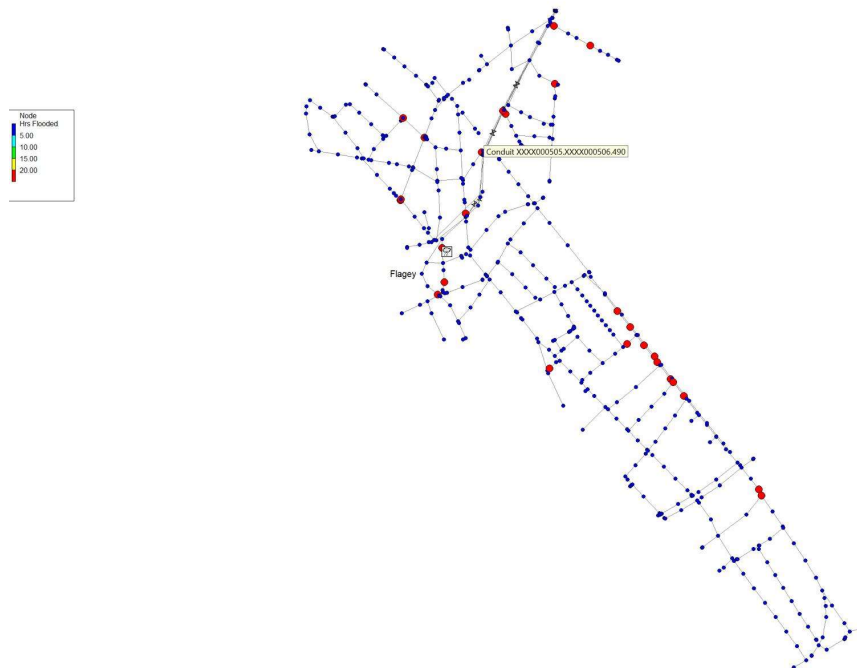


Figure 11: SWMM model showing node flooding at different locations

Figure 11 presents the simulation output specifically for the July 9th, 2024 storm, which was the first of the four major events modeled in SWMM. The red nodes represent

manholes that reached total surcharge and flooded during this specific event. While this is just one of the four scenarios simulated, it serves as a representative baseline for the network's vulnerability. The surcharges are most concentrated along the lower reaches of the collector, where the cumulative upstream volume exceeds the pipe's maximum capacity, forcing wastewater to reach the surface.

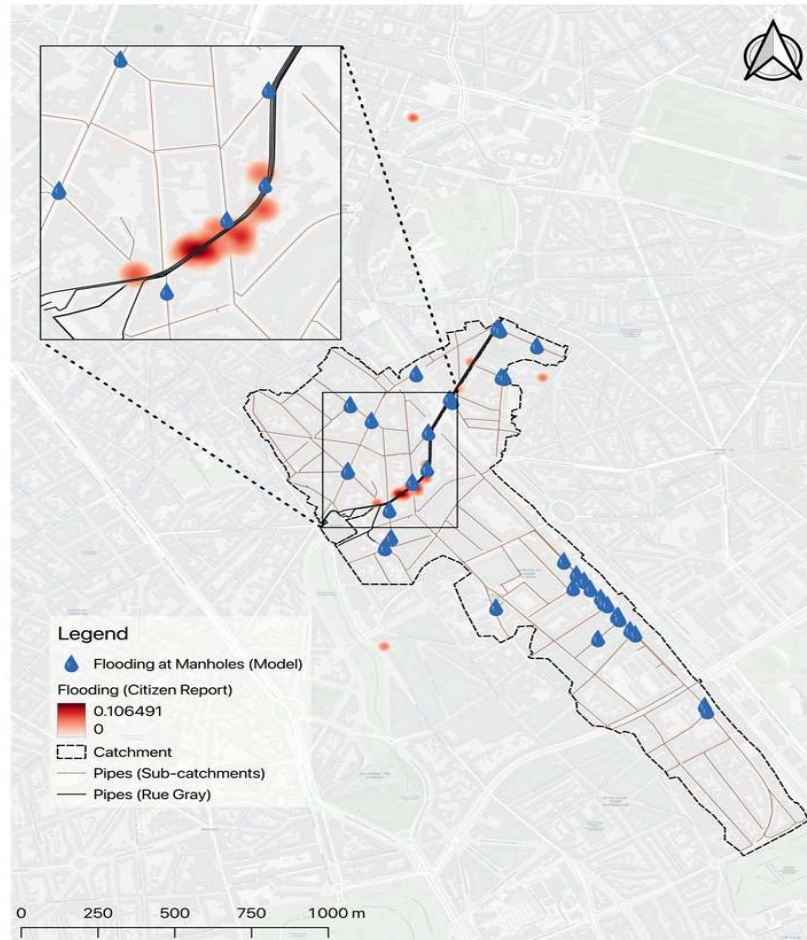


Figure 12: Validation of the SWMM model with citizen observations

The SWMM model was simulated using the rainfall event that occurred on 9 July 2024. This event was selected as a representative case to evaluate the model's performance under real conditions.

Following the simulation, the node flooding results were analyzed. The model identified multiple locations where flooding occurred within the drainage network.

These simulated flooding locations were then compared with citizen-reported observations from the same rainfall event. A strong agreement was found between the model results and the observed flood-prone areas.

This correspondence indicates that the model can reasonably represent the real system

behavior. Therefore, the model can be considered successfully validated for this event and is suitable for further analysis.

3.3 IDENTIFY HYDROLOGICAL FLUXES AND DYNAMICS

ObservationName	Measured	Modeled	Residual
Head_obs1_1	34.46	32.788383	1.6716166
Head_obs2_1	19.12	29.924303	-10.8043
Head_obs3_1	16.6	26.394952	-9.794952
Head_obs4_1	52.45	57.874008	-5.424008
Head_obs5_1	63.17	63.292044	-0.122044
Head_obs6_1	47.52	49.077338	-1.557338
Head_obs7_1	65.71	52.97263	12.73737
Head_obs8_1	48.33	46.081806	2.2481943
Head_obs9_1	53.38	50.347992	3.0320078
Head_obs10_1	52.12	49.151716	2.9682845
Head_obs11_1	51.86	48.210736	3.649264
Head_obs12_1	49.51	46.363478	3.1465222
Head_obs13_1	49.93	42.192161	7.7378386

Table 2: Results in observation points ModelMuse Run 5

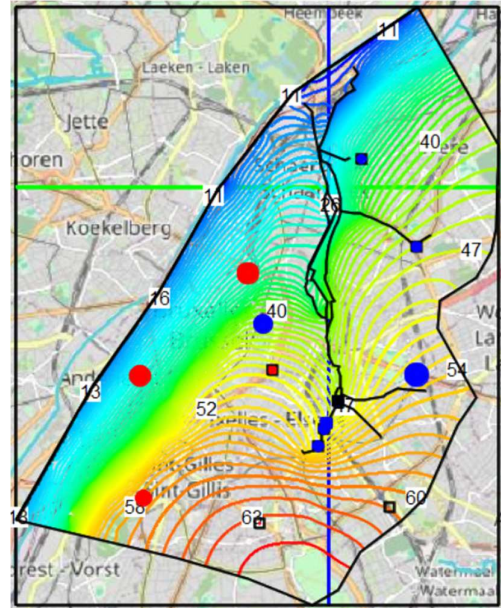


Figure 13: Output ModelMuse Run 5

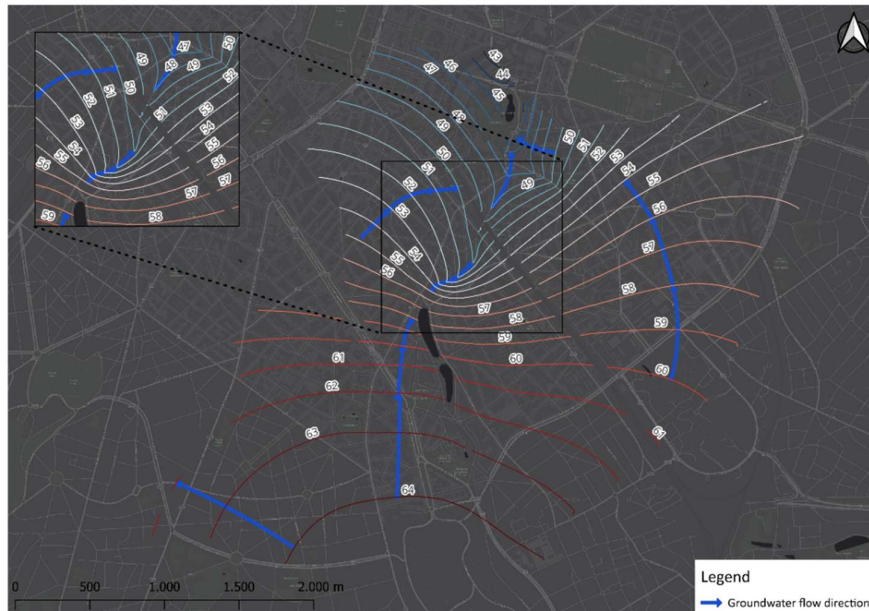


Figure 14: Contour map ModelMuse Run 5

The model's performance was evaluated by comparing predicted heads against 13 observation points, where the PEST calibration resulted in a significantly improved RMSE of 6.30 m, indicating a high level of reliability for the steady-state setup. As shown in Table 2, these numerical results are further visualized in the spatial validation of Figure 13, where residuals are represented by colored indicators. Within this mapping, blue dots indicate a modeling underestimation, signifying that observed water levels are higher than the model predicts, while red dots represent a modeling overestimation, where the model predicts higher heads than those actually measured. The specific concentration of blue dots near the valley bottom at Rue Gray suggests that even in a natural, barrier-free state, the actual groundwater levels remain slightly higher than the predicted steady-state regional levels, highlighting the area's inherent sensitivity to high water tables.

The contour map in Figure 14 illustrates the unhindered groundwater flow direction. In this scenario, the hydraulic heads follow the topographic gradient seamlessly toward the Senne River on the West. Without the "damming effect" of the railway, the contour lines exhibit a smoother distribution, representing the valley's natural drainage capacity. The results of Run 5 serve as the "control" state. By establishing this accurate steady-state baseline, we can isolate exactly how much the water table rises when the infrastructure barriers are reintroduced in the subsequent analysis of the "Damming Effect."

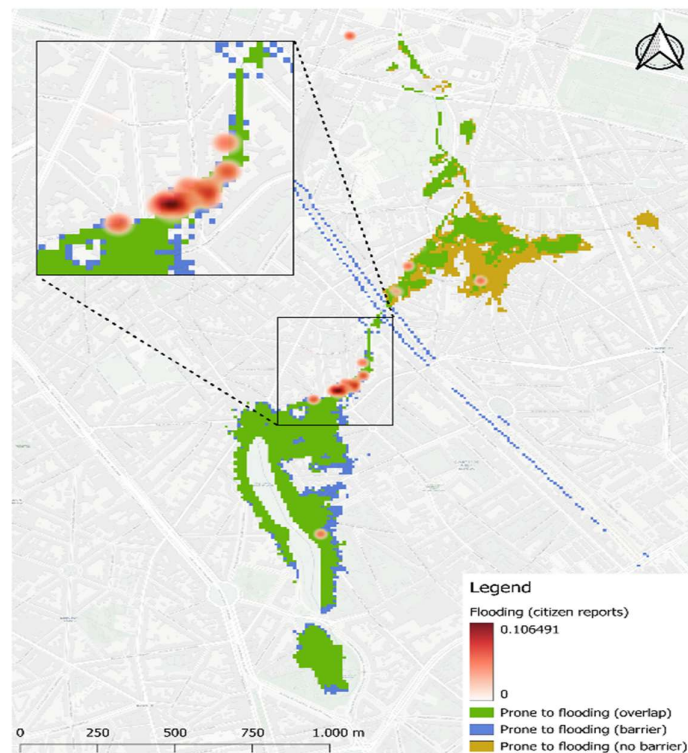


Figure 15: Map showing areas prone to flooding in the Maelbeek valley. Results based on ModelMuse outputs

To highlight the difference between the barriers and no barriers scenarios, a flood prone area map is created. Flood-prone areas are defined as places where the groundwater is

found at a maximum depth of four meters or less. Overlaying both the barriers and no barriers scenarios results in the following map (Figure 15). As can be seen, the barriers mainly result in more flood prone areas upstream of the barriers, whereas no barriers result in higher groundwater levels downstream of the barriers.

Apart from this difference, the map also shows the confirmation of the possibility of groundwater flooding in Rue Gray. When comparing the flood-prone areas with the citizen observations during the 9th of July 2024, it can be seen that all observations have been done in the area that is highlighted as flood-prone for both the barriers and no barriers scenarios. This means that even though the barriers have an influence, this area is prone to flooding anyways, confirming the expectations of groundwater intrusion in low lying areas like basements.

However, the steady-state approach in Run 3 produced unrealistic hydraulic heads during the peak intensity of the July 9th storm, indicating that the current single-layer configuration cannot fully absorb such high-intensity recharge. Furthermore, the model significantly underestimated sewer baseflow, with the simulated drainage of 8,299 m³/d falling far short of the calculated 38,000m³/d.

Moving forward, the research recommends a transition toward a more complex, multilayer geological model to better represent the stratification between topsoil and the "Sand of Brussels" aquifer. To achieve a more precise match between modeled and measured sewer outflows, future iterations must re-evaluate the recharge values used in the initial steady-state setup and correct the spatial placement of Observation Point 13. By addressing these structural and data-driven limitations, the model can evolve into a more robust diagnostic tool for managing the compound flood risks posed by both surface runoff and rising groundwater in the Brussels region.

4. CONCLUSION

This study successfully diagnosed the compound flooding mechanisms in the Rue Gray area by integrating citizen science with numerical hydraulic and hydrogeological models. The synthesis of citizen reports and SWMM simulations confirms that the primary driver of immediate surface flooding is the incapacity of the historic sewer network to manage high-intensity "cloudburst" events, such as that of July 9th, 2024. The rapid appearance of "brown water" at topographical low points highlights a critical infrastructure bottleneck where the drainage system reaches total surcharge almost instantaneously.

Simultaneously, the ModelMuse results reveal a secondary, subsurface dimension to the flood risk. The presence of anthropogenic barriers (the railway and bridge foundations) creates a significant "damming effect," artificially elevating the water table upstream. However, the consistent overlap of flood-prone zones and citizen observations in Rue Gray proves that the area is naturally vulnerable due to its valley-bottom topography. Even if surface drainage is improved, the shallow groundwater table remains a persistent threat to subterranean structures, particularly when the soil reaches saturation during extreme recharge events.

Ultimately, managing flood risk in the Maelbeek Valley requires a holistic approach that addresses both surface runoff and groundwater dynamics. While the current model provides a strong regional baseline, future efforts should transition toward transient, multi-layer simulations to better capture the time-dependent response of the aquifer to extreme weather. By correcting data spatiality and refining geological parameters, these models can serve as more precise tools for urban planning and the implementation of resilient drainage solutions in Brussels.

BIBLIOGRAPHY

- I. Goossens, S., Ortacesme, V., Ghekiere, P., Yilmaz, O., et al. (2014, November). *Transformation of Brussels' Urban Cultural Landscapes in the Case of Maalbek Valley: New Design Concepts* [Paper presentation]. The Fifth International Cultural Landscape Conference (INCUL 2014), Istanbul, Turkey.
- II. Bruxelles Environnement. (n.d.). *Carte du bassin versant du Maelbeek* . Datastore Brussels. <https://datastore.brussels/web/data/dataset/0be3a6d6-ce64-46a9-a2e3-822683f111f2>
- III. Hydria. (n.d.). *FlowBru: The hydrological monitoring network of the Brussels-Capital Region*. Retrieved from <https://hydria.be/flowbru/>
- IV. Bruxelles Environnement. (n.d.). *Brugeotool: Integrated management of the Brussels subsurface* .Retrieved from <https://geodata.environnement.brussels/client/brugeotool/>